

CSE332L Questions for Q&A of Report

LAB 1

1. What to connect to the enable pins of a MUX IC that has active-high enable, and why?
2. Why did we use a multiplexer, not a decoder, to build a 2-bit logic unit?
3. Design a 3-bit Arithmetic Unit that does the following operations:

S ₁	S ₀	C _{in}	A ₂	A ₁	A ₀	B ₂	B ₁	B ₀	D ₂	D ₁	D ₀	C _{out}	Microoperation
0	0	0	1	0	1	0	0	1					Subtract with Borrow
0	0	1	1	0	0	0	1	0					Subtract
0	1	0	1	1	1	0	1	1					Decrement A
0	1	1	1	1	0	0	0	1					Transfer A
1	0	0	1	0	1	0	1	1					Transfer A
1	0	1	1	1	0	1	1	1					Increment A
1	1	0	1	0	0	0	0	1					Add
1	1	1	0	0	1	1	0	1					Add with carry

- i. Draw the circuit diagram (Logisim) and the truth table. Explain the connections in your circuit.

LAB 2

1. Design a 5-bit Up counter with T Flip-flop in Logisim. Show the truth table.
2. Design a 5-bit Down counter with D Flip-flop in Logisim. Show the truth table.

LAB 3

Design a 5-bit Universal Shift Register that performs 4 operations:

S₁ S₀	Operations
0 0	Parallel Load (I ₅ I ₄ I ₃ I ₂ I ₁)
0 1	Circular Left Shift
1 0	Circular Right Shift
1 1	No Change

1. Draw the circuit diagram (Logisim) and the truth table.
2. What would happen to the output of the circuit if you connected GND to preset and VCC to clear? Explain with details.